Extraction of Visually Impaired Rights and Provisions from D4 PDF

Accessible Environment and Services for Visually Impaired Persons

- **Tactile Ground Surface Markers:** Accessible pathways must include tactile paving. *Guiding pavers* (300×300 mm tiles with raised parallel bars) help lead people with visual impairments along circulation routes. *Warning pavers* (300×300 mm tiles with raised blister domes) are installed at the edges of ramps, staircases and hazards to alert persons with visual impairments of approaching danger.
- **Ramps:** Ramps should have slip-resistant surfaces and be well illuminated. Tactile warning blocks must be placed at the top and bottom of each ramp (within 300–600 mm of the ramp ends) to warn visually impaired users. Ramps at buildings with limited space may use L-shaped or switchback designs.
- **Stairs:** All stair edges must be clearly visible. Step edges should have a **50-mm wide band of high-contrast color** (paint or tape) running the full width of each step, so that people with low vision can easily distinguish tread depth and riser height. Stairs must have warning tactile pavers (300 mm before and after each flight) to indicate the beginning and end of steps. Uniform riser heights (max 150 mm) and treads (min 300 mm) are required. Checklists emphasize that "each step edge [should be] visually highlighted" and that handrails be provided on both sides of every flight.
- **Handrails and Floor-Level Indicators:** Continuous handrails are required on both sides of ramps and stairs. Handrails should be circular (38–40 mm diameter) and mounted at two heights (760 mm and 900 mm above the floor). In multi-storey buildings, tactile markers (e.g. raised dots) are placed on the handrail ends to indicate the floor level: one dot for first floor, two for second, etc., to help a visually impaired person determine location.
- **Elevators (Lifts):** Elevators must be fully accessible. Control panels inside lifts should have **Braille labels and raised letters on high-contrast backgrounds**, so buttons can be identified by touch. Elevator doors should include vision panels at two levels (approximately 800 mm and 1500 mm above floor) so a person can see into/out of the car. Audio announcements of door opening/closing and floor levels must be provided for blind users. The interior of the car should minimize reflective surfaces (to avoid glare and confusion) and have a slip-resistant floor with no more than 10 mm gap at the doorway. Mirrors (if any) must be set at least 900 mm above the floor "to avoid confusing people with visual impairments". (Emergency communication systems are also equipped with induction loops for the hearing impaired.)
- **Doors and Entrances:** Entrances and doors should be easy to detect. Vision panels are required on doors between 800–1500 mm height, so a person can see if someone is on the other side. Doors must be **color-contrasted against the wall** (e.g. use a bright door against a light wall) to make them visible. Heavy or hard-to-operate doors are not allowed; ease of opening is emphasized (no more than 22 N force). Recessed or thin rubber floor mats or tactile strips should be placed at entrances to allow a visually impaired person to detect the doorway by foot. Accessible toilet doors are designed to swing outward (or slide) to leave enough space for assistance if needed.
- **Corridors and Floor Surfaces:** Corridors must be **wide and well lit**, with clear lines of sight to aid orientation. A strong color contrast is recommended between the floor and walls (and between walls and ceiling) to help people with low vision stay

oriented. One or more thick strips of bright color (e.g. yellow or orange) on the corridor floor can serve as a directional guide for the visually impaired. The floor finish itself should be uniform in shade (no busy patterns) to avoid confusion. Matt, anti-skid flooring is preferred; not only does it prevent slipping, it also provides a subtle tactile pattern underfoot that can aid navigation for those with vision loss. Corridors must be kept free of low-hanging obstacles (nothing protruding more than 100 mm) and maintain a minimum clear width (≥1500 mm) with turning spaces, so that guidance cues are unobstructed.

- **Switches and Controls:** Switchboards, controls and outlets should be visibly distinguishable. Light switches and fuse boxes must be outlined or painted in a contrasting color from the wall, making them stand out for someone with low vision. Switches should also have tactile markers: for example, a raised fan icon on the fan switch and a lightbulb icon on the light switch, so the switch's function can be felt. (All controls should be mounted between 400–1100 mm above floor to accommodate seated or shorter users.)
- **Toilet Facilities:** Restroom facilities must include visual cues and accessibility features. Signs for general (universal) toilets should use **bold, high-contrast pictograms**. For visually impaired users, each sign must be on a rigid plate with raised pictograms and **Braille or embossed lettering** (e.g. male symbol in triangle, female in circle, with Braille text). These signs are mounted beside each door at about 1400–1600 mm height for easy reach. Tactile warning pavers or textured mats should be installed 300 mm before and after toilet entrances to signal the doorway location. The toilet seat (WC) and adjacent wall surfaces should use contrasting colors so the fixtures stand out for low-vision users. All toilet rooms are equipped with visual fire alarms and emergency pull-switches (one near floor and one at about 900 mm height by the commode) that trigger an alarm. At least one urinal or washbasin in public toilets must have a tactile warning paver nearby to help a blind person locate it.
- **Signage and Wayfinding:** Signage throughout buildings must be designed for tactile reading and high visibility. Every permanent sign (room numbers, directional signs, emergency exit maps, etc.) should use **sans-serif fonts, large text and strong color contrast**, and should include Braille beneath the text. In particular, accessible toilet signs are standardized: a navy-blue plate with raised white letters (25 mm high) and 1 mm thick border, plus Braille directly below the text. Signs for elevators, stairwells and exits likewise must have raised letters and Braille on contrasting backgrounds. Maps and building directories should, whenever possible, incorporate a *tactile* map overlay or be available with audio guidance so that blind visitors can orient themselves. In multi-storey buildings, it is recommended that each floor's elevator and stairwell signage be color-coded and that stairwell walls have signs at landings indicating the current floor (aiding wayfinding in darkness or low vision).
- **Lighting:** Adequate, uniform lighting is vital for persons with low vision. All indoor activity areas should be brightly lit (250–300 lux recommended) to facilitate reading and navigation. Corridors and stairwells should have a minimum of 100 lux uniform illumination. Light fixtures should have shades or covers (no bare bulbs) to reduce glare and avoid flicker, creating a steady visual environment.
- **Communication and Information Access:** All information must be made accessible in multiple formats. Facilities should provide building information online (adhering to web accessibility standards) and display schematic maps with clear pictograms for each service area. Printed documents (notices, schedules, menus, tickets, etc.) must be available in **Braille, large-print, audio-recorded or simple-language versions**, in Hindi, English and any relevant local language. Electronic resources (websites, PDFs,

mobile apps) should be easily shared (e.g. via email or mobile devices) and support assistive technologies. Where forms or applications must be completed, users should be able to fill them electronically with accessible software, and staff assistance should be offered when needed. Personnel should be trained in basic sign language and visual communication; disability rooms and booths should be equipped with assistive devices such as **induction loops, audio orientation tools, and captioned video displays**. Staff must also be prepared to describe visual information verbally. Special care must be taken with lighting: areas used by visually impaired persons should have **uniform, non-glare lighting and no flicker**, so that low vision or hearing-impaired individuals are not disoriented by bright spots or flashing lights. Finally, an accessible feedback or grievance process should exist so that any person with a disability can lodge complaints or suggestions in an accommodating manner.

- **Sports and Recreation Facilities:** All sports venues and related buildings must include features for blind athletes and spectators. Competitions should provide **audio descriptions of the action** for visually impaired sports persons. Orientation materials (maps of the stadium or complex) should be provided in both visual and tactile formats, and downloadable wayfinding apps should include audio guidance. Information about events, rules or results must be available in Braille: for example, competition booklets and score sheets should be provided in Braille for blind athletes. Scoreboards and announcement systems are required to include audio output as well as visual displays, so that a blind spectator can follow the score. Medal ceremonies and podiums must be fully accessible: podiums should have ramps or steps with handrails and tactile indicators, and audio announcements should accompany any on-screen graphics. Locker rooms and equipment areas should have **tactile or large-print signage**: locker numbers must be embossed and supplemented with Braille, and banks of lockers should include some orientation system (e.g. tactile floor paths or landmarks). Surfaces in sports halls should avoid gloss or glare (use matte finishes and clear color contrast) to help all athletes, including those with low vision, distinguish areas and equipment. Finally, all accessible sports complexes are required by government mandate to meet these standards, ensuring that visually impaired athletes have equal access to training and competition facilities.

Sources: Government of India accessibility guidelines for persons with disabilities (compiled from the *Rights of Persons with Disabilities Act* notifications and the *Accessible Sports Complex Guidelines*).